Stochastic Calculus and Option Pricing
Synthesis Exercise Sheet

1 Week 1 — Option Pricing and Core Probability Tools

Part I — Random Variables and Distributions

Exercise 1.1 (Level 1 — A financial payoff as a random variable)**.** At maturity $T = 1$, a stock price has the physical distribution

s	70	105	140
$\mathbb{P}(S_T = s)$	0.25	0.50	0.25

and $X = (S_T - 100)^+$ is the payoff of a European call.

(a) Identify the underlying, maturity, strike, and possible values of X.

(b) Determine the law and the cdf of X.

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$.

(d) If $r = 3\%$ is continuously compounded, compute $e^{-rT}\mathbb{E}[X]$. Explain why this discounted physical expectation is not, by itself, a no-arbitrage price.

Exercise 1.2 (Level 1 — Cdf, density, and moments)**.** Let $U \sim \mathrm{Unif}[-1, 3]$.

(a) Write the cdf F_U on the whole real line and recover the density f_U.

(b) Compute $\mathbb{P}(0 < U \leq 2)$ from the cdf and again from the density.

(c) Starting from the defining integrals, compute $\mathbb{E}[U]$, $\mathbb{E}[U^2]$, and $\mathrm{Var}(U)$.

(d) Set $V = 2U + 1$. Find its support, cdf, mean, and variance.

Exercise 1.3 (Level 2 — Empirical moments, LLN, and CLT)**.** The observed daily claim counts in a short sample are

$$2,\ 5,\ 4,\ 3,\ 6.$$

In the population, suppose daily claim counts $X_1, X_2, \ldots$ are i.i.d. Poisson with parameter 4.

(a) Compute the sample mean, the empirical variance with denominator n, and the unbiased sample variance with denominator $n - 1$.

(b) State the LLN for $\overline{X}_n$ and interpret it here.

(c) State the CLT for $\overline{X}_n$. What are the approximate mean and standard deviation of $\overline{X}_{100}$?

(d) Use the CLT to approximate $\mathbb{P}(\overline{X}_{100} > 4.3)$. You may use $\Phi(1.5) \approx 0.9332$.

(e) Explain in one sentence the different conclusions supplied by the LLN and the CLT.

Exercise 1.4 (Level 2 — Characteristic function of a Gaussian)**.** Let $Z \sim \mathcal{N}(0,1)$ and define

$$\varphi_Z(u) = \mathbb{E}[e^{iuZ}], \qquad u \in \mathbb{R}.$$

The following steps derive the characteristic function without using a complex change of variables.

(a) Using the standard Gaussian density, write $\varphi_Z(u)$ as an integral and verify that $\varphi_Z(0) = 1$.

(b) Differentiate under the integral sign to express $\varphi_Z'(u)$. You may use

$$\frac{\mathrm{d}}{\mathrm{d}z} e^{-z^2/2} = -z e^{-z^2/2}.$$

(c) Integrate by parts and show that

$$\varphi_Z'(u) = -u\varphi_Z(u).$$

Check explicitly that the boundary term vanishes.

(d) Solve this differential equation using the initial condition from part (a), and obtain $\varphi_Z(u)$.

(e) If $X = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$, deduce that

$$\varphi_X(u) = \exp\!\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right).$$

(f) Let X_1 and X_2 be independent Gaussian variables. Use characteristic functions to identify the distribution of $X_1 + X_2$.

Part II — Joint Distributions and Information

Exercise 1.5 (Level 1 — Reading a joint law)**.** The joint law of (X, Y) is

	$Y = L$	$Y = H$	$\mathbb{P}(X = x)$
$X = 0$	0.10	0.40	
$X = 1$	0.20	0.30	
$\mathbb{P}(Y = y)$			1

(a) Complete the marginal probabilities in the table.

(b) Are X and Y independent? Give a numerical justification.

(c) Compute $\mathbb{P}(X = 1 \mid Y = L)$ and $\mathbb{P}(X = 1 \mid Y = H)$.

(d) Interpret X as a loss indicator and Y as a rating signal. Which signal carries the larger conditional loss probability?

> For a signal Y, $\sigma(Y)$ denotes exactly the information revealed by observing Y. On a finite space, its events are the unions of the level sets of Y.

Exercise 1.6 (Level 2 — Information generated by a signal)**.** A fair die is rolled and $Y = \mathbf{1}_{\{4,5,6\}}$. Let $\mathcal{G} = \sigma(Y)$.

(a) Give the partition of $\Omega = \{1, \ldots, 6\}$ generated by Y and list all events in $\mathcal{G}$.

(b) Decide whether each variable is $\mathcal{G}$-measurable:

$$Z_1 = Y, \qquad Z_2 = (-1)^Y, \qquad Z_3 = \mathbf{1}_{\{5,6\}}, \qquad Z_4 = 2 + 3Y.$$

Justify each answer using constancy on the atoms of the partition.

(c) Find $g : \{0, 1\} \to \mathbb{R}$ such that $Z_2 = g(Y)$.

(d) Explain what information is missing when one observes Y but wants to determine Z_3.

Part III — Conditional Expectation

Exercise 1.7 (Level 1 — Conditional expectation on a finite partition)**.** A fair die is rolled. The available information reveals only the block

$$L = \{1, 2\}, \qquad M = \{3, 4\}, \qquad H = \{5, 6\}, \qquad \mathcal{G} = \sigma(L, M, H).$$

Define $X = 6\mathbf{1}_{\{\omega \geq 4\}}$.

(a) Is X $\mathcal{G}$-measurable? Explain.

(b) Compute $\mathbb{E}[X]$ and the average of X on each atom L, M, H.

(c) Write $\mathbb{E}[X \mid \mathcal{G}]$ using indicator functions.

(d) Verify the defining averaging property on L, M, and H.

(e) Verify $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.

Exercise 1.8 (Level 2 — Conditional density and conditional moments)**.** The pair (X, Y) has joint density

$$f_{X,Y}(x, y) = (x + y)\mathbf{1}_{\{0<x<1,\ 0<y<1\}}.$$

(a) Verify that the density integrates to one and compute both marginal densities.

(b) For $0 < y < 1$, derive $f_{X|Y}(x \mid y)$ and its support. Are X and Y independent?

(c) Compute $\mathbb{E}[X \mid Y = y]$, then write $\mathbb{E}[X \mid Y]$ as a random variable.

(d) For $f(x) = x^2$, compute $\mathbb{E}[f(X) \mid Y = y]$. Identify g such that $\mathbb{E}[f(X) \mid Y] = g(Y)$ and explain why this is $\sigma(Y)$-measurable.

(e) Verify the tower property by comparing $\mathbb{E}[\mathbb{E}[X \mid Y]]$ with $\mathbb{E}[X]$.

Exercise 1.9 (Extension — Gaussian conditioning)**.** Let (X, Y) be jointly Gaussian with

$$\mathbb{E}[X] = \mu_X, \quad \mathbb{E}[Y] = \mu_Y, \quad \mathrm{Var}(X) = \sigma_X^2, \quad \mathrm{Var}(Y) = \sigma_Y^2 > 0,$$

and $\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.

(a) Set $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ and $R = X - \mu_X - a(Y - \mu_Y)$. Show that $\mathrm{Cov}(R, Y) = 0$.

(b) Use joint Gaussianity to conclude that R and Y are independent.

(c) Deduce formulas for $\mathbb{E}[X \mid Y]$ and $\mathrm{Var}(X \mid Y)$.

(d) Interpret $|\rho|$ as hedge effectiveness and the conditional variance as residual risk.

Part IV — Stochastic Processes and Filtrations

Exercise 1.10 (Level 2 — Three coin tosses viewed through time)**.** Three fair coins are tossed. Write $\omega = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{H, T\}^3$ and define

$$\mathcal{F}_0 = \{\varnothing, \Omega\}, \qquad \mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n), \quad n = 1, 2, 3.$$

Let S_n be the number of heads observed by time n.

(a) Describe the atoms of $\mathcal{F}_0, \ldots, \mathcal{F}_3$ and verify that these sigma-algebras form a filtration.

(b) Show that (S_0, S_1, S_2, S_3) is adapted.

(c) Define $A_0 = A_1 = 0$, $A_2 = \mathbf{1}_{\{\varepsilon_3 = H\}}$, and $A_3 = A_2$. At which time does (A_n) fail to be adapted? Interpret the failure.

(d) Let τ be the first head, capped at 3. Determine τ on every outcome and verify $\{\tau \le n\} \in \mathcal{F}_n$.

Part V — Martingales and Fair Games

Exercise 1.11 (Level 2 — A fair but asymmetric random walk)**.** Let $(\xi_k)_{k \ge 1}$ be independent with

$$\mathbb{P}(\xi_k = 2) = \frac{1}{3}, \qquad \mathbb{P}(\xi_k = -1) = \frac{2}{3}, \qquad S_n = \sum_{k=1}^{n} \xi_k, \qquad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n).$$

(a) Compute $\mathbb{E}[\xi_k]$ and $\mathbb{E}[\xi_k^2]$. Why is this a fair game?

(b) Prove from the definition that (S_n) is a martingale.

(c) Prove that $S_n^2 - 2n$ is a martingale.

(d) If instead $\mathbb{P}(\xi_k = 2) = 1/4$, find the predictable drift and construct a centered martingale from S_n.

(e) Explain the casino interpretation of the martingale property.

Exercise 1.12 (Level 2 — Replication and risk-neutral pricing)**.** A stock has $S_0 = 50$ and, after one period,

$$S_1 = 65 \quad \text{or} \quad S_1 = 45.$$

The risk-free gross return is $R = 1.05$. A call has strike $K = 55$ and payoff $H = (S_1 - K)^+$.

(a) Build the state-by-state payoff table.

(b) Find the risk-neutral up probability q from $S_0 = R^{-1}\mathbb{E}^q[S_1]$ and check that $q \in (0, 1)$.

(c) Compute $V_0 = R^{-1}\mathbb{E}^q[H]$.

(d) Find Δ shares and a maturity cash position B_1 such that $\Delta S_1 + B_1 = H$ in both states.

(e) Check that $\Delta S_0 + B_1/R = V_0$ and explain why the physical up probability does not determine this price.

(f) Relate the pricing identity to the martingale property of discounted asset prices under the risk-neutral probability.

Part VI — Mathematical Toolbox

Exercise 1.13 (Levels 1–2 — Derivatives, gradients, and the chain rule)**. (a)** Starting from the difference quotient, compute the derivative of $g(x) = x^2$.

(b) For $f(x, y) = x^2y + e^y$, compute all first and second partial derivatives, the gradient, and the Hessian.

(c) Let S be differentiable and let $V = V(t, s)$ be smooth. If $h(t) = V(t, S(t))$, derive $h'(t)$ from the multivariable chain rule.

(d) Apply the previous result to $V(t, s) = e^{-rt}s^2$.

(e) The payoff $v(s) = (s - K)^+$ is continuous but not differentiable at one point. Identify it and compute the derivative everywhere else.

2 Week 2 — Brownian Motion, Stopping Times, and Quadratic Variation

Part I — Gaussian Structure and Brownian Transformations

Exercise 2.1 (Level 2 — Gaussian vectors and the covariance kernel)**.** Let $(W_t)_{t\geq 0}$ be a standard Brownian motion and fix $0 < r < s < t$.

(a) Find the joint law of the increment vector

$$(W_r,\, W_s - W_r,\, W_t - W_s).$$

(b) Deduce that (W_r, W_s, W_t) is Gaussian and compute its covariance matrix.

(c) Compute the conditional distribution of W_t given $\mathcal{F}_s$, then given $W_s = x$.

(d) Show that the Brownian covariance kernel $K(u, v) = \min(u, v)$ is positive semidefinite, meaning that

$$\sum_{i,j=1}^{n} a_i a_j K(t_i, t_j) \geq 0$$

for all dates $t_1, \ldots, t_n$ and coefficients $a_1, \ldots, a_n$.

(e) Explain why the mean function and the covariance kernel determine every finite-dimensional Brownian law.

Exercise 2.2 (Level 2 — Symmetry, scaling, renewal, and time reversal)**.** Fix $c > 0$ and $s > 0$. Consider

$$X_t = -W_t, \qquad Y_t = \frac{W_{ct}}{\sqrt{c}}, \qquad Z_t = W_{s+t} - W_s, qquadt \geq 0,$$

and $R_t = W_{s-t} - W_s$ for $0 \leq t \leq s$.

(a) Verify from the defining properties that X and Y are standard Brownian motions.

(b) Show that Z is a standard Brownian motion independent of $\mathcal{F}_s$. Interpret this as deterministic-time renewal.

(c) Show that R has the same finite-dimensional laws as Brownian motion restricted to $[0, s]$.

(d) Deduce the square-root-of-time identity

$$\max_{0\le u\le ct} W_u \overset{d}{=} \sqrt{c}\max_{0\le u\le t} W_u.$$

(e) State the almost-sure lim sup and lim inf identities from the law of the iterated logarithm, both as $t\to\infty$ and as $t\downarrow 0$.

Part II — Markov Property, Martingales, and Stopping

Exercise 2.3 (Level 2 — Markov semigroup and martingale correction)**.** For bounded measurable f and $h>0$, define

$$(P_h f)(x) = \int_{\mathbb{R}} f(x+y)\frac{e^{-y^2/(2h)}}{\sqrt{2\pi h}}\,\mathrm{d}y.$$

(a) Prove the conditional-expectation form of the Markov property:

$$\mathbb{E}[f(W_{t+h}) \mid \mathcal{F}_t] = (P_h f)(W_t).$$

(b) Prove the semigroup identity $P_{h+k} = P_h P_k$.

(c) For $0\le s<t$, compute $\mathbb{E}[W_t \mid \mathcal{F}_s]$ and $\mathbb{E}[W_t^2 \mid \mathcal{F}_s]$.

(d) Deduce directly that W_t and $W_t^2 - t$ are martingales.

(e) State the general proposition defining $\langle X\rangle$ for a continuous square-integrable martingale X, and identify $\langle W\rangle_t$.

Exercise 2.4 (Level 2 — Stopping times and information at stopping)**.** Fix $T>0$, $a>0$, and $\ell<0<u$. Consider

$$\tau_a = \inf\{t\ge 0 : W_t = a\}, \qquad \tau_{\ell,u} = \inf\{t\ge 0 : W_t\notin(\ell,u)\},$$

$$\rho_T = \sup\{0\le t\le T : W_t = 0\}, \qquad \gamma_T = \operatorname*{arg\,max}_{0\le t\le T} W_t.$$

(a) Determine which of $T, \tau_a, \tau_{\ell,u}, \rho_T, \gamma_T$ are stopping times and justify each answer.

(b) Prove, using path continuity, that

$$\{\tau_a\le t\} = \left\{\max_{0\le v\le t} W_v \ge a\right\}.$$

(c) Give the definition of $\mathcal{F}_\tau$. Explain why W_τ is $\mathcal{F}_\tau$-measurable for an almost surely finite stopping time τ.

Part III — Quadratic Variation

Exercise 2.5 (Level 2 — Realized quadratic variation)**.** Fix $T>0$, let $t_i = iT/n$, and define

$$Q_n = \sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})^2.$$

(a) Compute $\mathbb{E}[Q_n]$ and $\mathrm{Var}(Q_n)$.

(b) Deduce that $Q_n \to T$ in L^2.

(c) Let A be continuously differentiable. Prove that $[A]_T = 0$.

(d) Show that the cross-term

$$\sum_i (A_{t_{i+1}} - A_{t_i})(W_{t_{i+1}} - W_{t_i})$$

converges to zero in probability.

(e) Deduce that, for $X_t = A_t + \sigma W_t$, $[X]_T = \sigma^2 T$.

(f) If X is observed on the grid, propose a consistent estimator of σ^2 based on its squared increments.

Part IV — Price Models and Trading Gains

Exercise 2.6 (Level 2 — Symbolic analysis of geometric Brownian motion)**.** Let $S_0 > 0$, $\sigma > 0$, and

$$S_t = S_0 \exp\Big(\Big(\mu - \tfrac{1}{2}\sigma^2\Big)\, t + \sigma W_t\Big).$$

(a) Find the law of $\log(S_t/S_0)$ and the conditional law of $\log(S_t/S_s)$ given $\mathcal{F}_s$, for $0 \le s < t$.

(b) Compute $\mathbb{E}[S_t]$, $\mathrm{Var}(S_t)$, and the median of S_t.

(c) For a level $K > 0$, express $\mathbb{P}(S_t > K)$ using Φ.

(d) Compute $[\log S]_t$ and explain why the drift does not appear.

(e) For a partition of $[0, T]$, show how squared log-returns provide a consistent estimator of σ^2 as the mesh tends to zero.

Exercise 2.7 (Level 2 — A two-step adapted trading gain)**.** Fix $0 < t_1 < T$. Let $\Delta_0 \in \mathbb{R}$ and let Δ_1 be a square-integrable $\mathcal{F}_{t_1}$-measurable random variable. Define

$$G_T = \Delta_0 W_{t_1} + \Delta_1 (W_T - W_{t_1}).$$

(a) Explain why the measurability of Δ_1 is the no-look-ahead condition.

(b) Prove that $\mathbb{E}[G_T] = 0$.

(c) Show that the two gain terms are orthogonal in L^2, and deduce

$$\mathbb{E}[G_T^2] = \Delta_0^2 t_1 + (T - t_1)\mathbb{E}[\Delta_1^2].$$

(d) Explain why choosing $\Delta_1 = \mathbf{1}_{\{W_T - W_{t_1} > 0\}}$ is inadmissible.

(e) Write the corresponding piecewise constant process Δ_t and express G_T as $\int_0^T \Delta_t \, \mathrm{d}W_t$.

3 Week 3 — Applications of Stochastic Integration

Throughout, W is standard Brownian motion with its usual augmented natural filtration $(\mathcal{F}_t)$, and $T > 0$ is deterministic.

Part I — Information and Identification

Exercise 3.1 (Easy — Uncorrelated gains can remain dependent)**.** Fix $0 < a < T$. Consider the successive gains

$$U = \int_0^a 1 \, \mathrm{d}W_t, \qquad V = \int_a^T W_a \, \mathrm{d}W_t.$$

(a) Explain why the position defining V is predictable and has finite expected energy. Express U and V using Brownian increments.

(b) Compute $\mathbb{E}[V \mid \mathcal{F}_a]$ and $\mathbb{E}[V^2 \mid \mathcal{F}_a]$. Deduce $\mathbb{E}[V]$ and $\mathbb{E}[V^2]$.

(c) Show that $\mathrm{Cov}(U, V) = 0$.

(d) Are U and V independent? Justify your answer by examining $\mathbb{E}[V^2 \mid U]$. Explain why disjoint holding intervals do not settle the question when the later position depends on the earlier gain.

Exercise 3.2 (Easy — Can the terminal gain identify the strategy?)**.** Let ϕ, ψ be predictable with $\mathbb{E}\int_0^T (\phi_t^2 + \psi_t^2) \, \mathrm{d}t < \infty$, and write $I(\phi) = \int_0^T \phi_t \, \mathrm{d}W_t$.

(a) Suppose $I(\phi) = I(\psi)$ almost surely. Use the isometry to prove that $\phi = \psi$ almost everywhere for the product measure $\mathrm{d}t \otimes \mathbb{P}$.

(b) Does equality in *distribution* of the two gains imply the same conclusion? Construct a counterexample with constant integrands.

(c) Suppose $I(\phi) \geq 0$ almost surely. Show first that $I(\phi) = 0$ almost surely, then identify ϕ up to $\mathrm{d}t \otimes \mathbb{P}$-null sets.

(d) Explain why changing a deterministic integrand at a single deterministic time cannot be detected from its stochastic integral.

Part II — What Makes an Approximation Valid?

Exercise 3.3 (Easy — An exposure that becomes singular at maturity)**.** For $\alpha \geq 0$, consider the deterministic function $f_\alpha(t) = (T-t)^{-\alpha}$ for $0 \leq t < T$; its value at T may be set to zero. For $0 < \varepsilon < T$, define

$$I_\varepsilon = \int_0^{T-\varepsilon} f_\alpha(t)\,\mathrm{d}W_t.$$

(a) Compute $\mathbb{E}[I_\varepsilon]$ and $\mathbb{E}[I_\varepsilon^2]$, treating $\alpha = 1/2$ separately.

(b) Determine exactly when $f_\alpha \in L^2(0,T)$, so that the full-horizon integral I exists under the construction of Lecture 3.

(c) In that case, compute the exact mean-square truncation error $\mathbb{E}[(I - I_\varepsilon)^2]$ and show that it tends to zero.

(d) For the remaining values of α, explain why I_ε cannot converge in L^2 as $\varepsilon \downarrow 0$, although each truncated integral is well defined.

Exercise 3.4 (Easy to medium — Pointwise convergence can lose the risk)**.** On $[0,1]$, let

$$f_n(t) = \sqrt{n}\,\mathbf{1}_{(0,1/n]}(t), \qquad J_n = \int_0^1 f_n(t)\,\mathrm{d}W_t.$$

(a) Show that $f_n(t) \to 0$ for every $t \in [0,1]$, but compute $\int_0^1 f_n(t)^2\,\mathrm{d}t$. Does $f_n \to 0$ in $L^2(0,1)$?

(b) Express J_n as a Brownian value and identify its law. Can J_n converge to zero in probability?

(c) Compute $\mathbb{E}[J_n J_{2n}]$ and $\mathbb{E}[(J_n - J_{2n})^2]$. Deduce that (J_n) has no L^2 limit.

(d) Replace $\sqrt{n}$ by n^γ, with $\gamma \in \mathbb{R}$. For which γ do the corresponding stochastic integrals converge to zero in L^2? State the lesson for interchanging limits and stochastic integrals.

Part III — Choosing and Approximating an Exposure

Exercise 3.5 (Medium — The least risky deterministic exposure)**.** Fix $c \in \mathbb{R}$. Among deterministic functions $f \in L^2(0,T)$ satisfying $\int_0^T f(t)\,\mathrm{d}t = c$, consider the gain $G_f = \int_0^T f(t)\,\mathrm{d}W_t$.

(a) Express $\mathrm{Var}(G_f)$ in terms of f. Use Cauchy–Schwarz to obtain a lower bound depending only on c, T.

(b) Find an exposure attaining this bound, and prove uniqueness up to equality almost everywhere.

(c) Suppose trading is allowed only during $(a, T]$, where $0 < a < T$. Find the new minimizer and minimum variance under the same constraint $\int_0^T f(t)\,\mathrm{d}t = c$.

(d) For any admissible f in the original problem, compute $\mathbb{E}[(G_f - (c/T)W_T)^2]$. Interpret the result as the additional variance caused by an uneven exposure profile.

Exercise 3.6 (Medium — The best position held between two observations)**.** Fix $0 \le a < b$. Approximate

$$I = \int_a^b W_t\,\mathrm{d}W_t \quad \text{by} \quad J_\xi = \xi(W_b - W_a),$$

where $\xi \in L^2(\Omega, \mathcal{F}_a, \mathbb{P})$ is chosen at time a.

(a) Use the isometry to express $\mathbb{E}[(I - J_\xi)^2]$ as a time integral. Verify that all integrands have finite expected energy.

(b) Write $W_t - \xi = (W_a - \xi) + (W_t - W_a)$, and use conditioning on $\mathcal{F}_a$ to show that

$$\mathbb{E}[(I - J_\xi)^2] = (b-a)\mathbb{E}[(W_a - \xi)^2] + \frac{(b-a)^2}{2}.$$

(c) Find the optimal ξ, its uniqueness condition, and the smallest possible error. Explain why a positive error remains.

(d) On a deterministic grid $0 = t_0 < \cdots < t_n = T$, deduce the minimum error when one position is chosen at each t_i and held on $(t_i, t_{i+1}]$. For fixed n, show that the uniform grid is optimal. *Hint:* use $\sum_i (t_{i+1} - t_i) = T$ and Cauchy–Schwarz.

Part IV — Accumulated Risk and Risk Limits

Exercise 3.7 (Medium — Integrating a stochastic gain again)**.** Define

$$M_t = \int_0^t s \, \mathrm{d}W_s, \qquad N_t = \int_0^t M_s \, \mathrm{d}W_s.$$

You may use the covariance form of the isometry from Lecture 3.

(a) Compute $\mathbb{E}[M_t]$ and $\mathbb{E}[M_t^2]$. Check that M is an admissible integrand for N on every bounded horizon.

(b) Prove that N is a square-integrable martingale, and compute $\mathbb{E}[N_t^2]$ without finding an explicit formula for N_t.

(c) Compute $\mathrm{Cov}(N_t, W_t)$ and $\mathrm{Cov}(N_t, M_t)$. Explain why these calculations alone do not establish independence.

(d) For $0 \le s < t$, find $\mathbb{E}[N_t \mid \mathcal{F}_s]$ and show that

$$\mathbb{E}[(N_t - N_s)^2 \mid \mathcal{F}_s] = (t-s)M_s^2 + \int_s^t \int_s^u r^2 \, \mathrm{d}r \, \mathrm{d}u.$$

Hint: first compute $\mathbb{E}[M_u^2 \mid \mathcal{F}_s]$.

Exercise 3.8 (Medium — Stopping when a variance budget is spent)**.** Let ϕ be predictable with $\mathbb{E}\int_0^T \phi_s^2 \, \mathrm{d}s < \infty$. For $b > 0$, define

$$A_t = \int_0^t \phi_s^2 \, \mathrm{d}s, \qquad \tau = \inf\{t \in [0,T] : A_t \ge b\} \wedge T,$$

with $\inf \varnothing = \infty$, and let $G_t = \int_0^t \mathbf{1}_{\{s \le \tau\}} \phi_s \, \mathrm{d}W_s$.

(a) Using continuity, adaptation, and monotonicity of A, show that τ is a stopping time. Explain why the stopped integrand is predictable.

(b) Show pathwise that $\int_0^T \mathbf{1}_{\{s \le \tau\}} \phi_s^2 \, \mathrm{d}s \le b$. Deduce $\mathbb{E}[G_T] = 0$, $[G]_T \le b$, and $\mathbb{E}[G_T^2] \le b$.

(c) For $L > 0$, prove $\mathbb{P}(|G_T| \ge L) \le b/L^2$. Does this imply the pathwise loss guarantee $G_T \ge -L$?

(d) Set $\phi_t = \sqrt{b/T}$. Identify τ and the law of G_T, and use this example to settle the preceding question.

4 Week 4 — Theoretical Applications of the Itô Formula

Unless otherwise specified, W is standard Brownian motion, all initial values are deterministic, and coefficients are constant. Work on bounded time horizons and justify each use of a zero-mean stochastic integral.

Part I — Transformations and Their Meaning

Exercise 4.1 (Easy — A transform whose curvature changes sign)**.** Let $Y_t = \log(1 + W_t^2)$.

(a) For $f(x) = \log(1 + x^2)$, compute $f'(x)$ and $f''(x)$. Explain why the classical Itô formula applies on the whole real line.

(b) Find the drift and diffusion coefficient of Y. Determine the regions where its drift is positive, zero, or negative.

(c) Find an explicit function a such that

$$Y_t - \int_0^t a(W_s)\,\mathrm{d}s$$

is a true martingale. Check this by bounding the diffusion coefficient.

(d) Compute $[Y]_t$ as an integral. Explain why the sign changes in the drift do not produce negative increments of quadratic variation.

Exercise 4.2 (Easy to medium — Recovering the noise by an inverse transform)**.** Set $X_t = \sinh(W_t)$, and recall that

$$F(x) = \operatorname{arsinh}(x) = \log\bigl(x + \sqrt{1 + x^2}\bigr).$$

(a) Apply Itô to $\sinh(W_t)$ and express both coefficients entirely in terms of X_t. Use $\cosh^2 w - \sinh^2 w = 1$.

(b) Compute $F'(x)$ and $F''(x)$, then apply Itô to $F(X_t)$. Check explicitly that the drift cancels and the diffusion coefficient is one.

(c) A student keeps only $F'(X_t)\,\mathrm{d}X_t$. Compute the spurious drift produced by that calculation.

(d) Now let $\mathrm{d}Z_t = a(Z_t)\,\mathrm{d}t + \sqrt{1 + Z_t^2}\,\mathrm{d}W_t$. Determine the drift function $a(x)$ for which the drift of $F(Z_t)$ vanishes identically. In that case, identify $F(Z_t)$ from its initial value and differential.

Part II — Which Martingale Claims Survive?

Exercise 4.3 (Easy to medium — The same transform under two dynamics)**.** Let $f \in C^2(\mathbb{R})$ and fix $\mu \neq 0$. Consider, for every starting point $x \in \mathbb{R}$,

$$X_t^x = x + W_t, \qquad Y_t^x = x + \mu t + W_t.$$

(a) Write the drift coefficient of $f(X_t^x)$ and of $f(Y_t^x)$.

(b) Determine all f for which the drift of $f(X_t^x)$ vanishes identically for every x.

(c) Among those functions, determine which also give identically zero drift for $f(Y_t^x)$. Explain why a martingale conclusion depends on the dynamics of the input process.

(d) Verify directly that the candidate processes in parts (b) and (c) are true martingales on bounded horizons, rather than stopping at a formal zero-drift calculation.

Exercise 4.4 (Medium — Correlation between two compensated squares)**.** Let (W, B) be jointly Brownian with $\mathrm{d}[W, B]_t = \rho\,\mathrm{d}t$, $-1 \leq \rho \leq 1$, and set

$$M_t = W_t^2 - t, \qquad N_t = B_t^2 - t.$$

You may use $\mathbb{E}[W_t B_t] = \rho t$ and the fact that Gaussian variables have finite moments of every order.

(a) Find $\mathrm{d}M_t$, $\mathrm{d}N_t$, and $\mathrm{d}[M, N]_t$.

(b) Apply the product rule to $M_t N_t$. Find the finite-variation term that must be subtracted to obtain a true martingale. Justify the integrability of its stochastic terms using Gaussian moments.

(c) Take expectations to compute $\mathbb{E}[M_t N_t]$, then find the correlation of M_t and N_t for $t > 0$. *Hint:* each compensated square has variance $2t^2$.

(d) Explain why changing ρ to $-\rho$ leaves this correlation unchanged. For which ρ is MN itself a martingale? Check the cases $\rho = 0$ and $|\rho| = 1$.

Part III — Products with Accumulated Quantities

Exercise 4.5 (Medium — A random discount factor has finite variation)**.** Define

$$A_t = \int_0^t W_s^2 \, \mathrm{d}s, \qquad D_t = e^{-A_t}, \qquad Z_t = D_t W_t.$$

(a) Find $\mathrm{d}A_t$ and $\mathrm{d}D_t$. Explain why both processes have finite variation on bounded intervals and why $0 < D_t \leq 1$.

(b) Compute $[D, W]_t$ and apply the product rule to Z_t. Explain why randomness of D does not create a mixed correction here.

(c) Find h_t explicitly so that

$$K_t = Z_t + \int_0^t h_s \, \mathrm{d}s$$

is a true martingale starting at zero. Prove the required expected-energy bound for its stochastic integrand.

(d) Compute $[K]_t$ and show that $\mathbb{E}[K_t^2] \leq t$. Does adding back the finite-variation term change quadratic variation?

Exercise 4.6 (Medium — A geometric model with a continuous inflow)**.** Suppose an account receives cash at a constant rate $c > 0$ and satisfies

$$\mathrm{d}X_t = (\mu X_t + c) \, \mathrm{d}t + \sigma X_t \, \mathrm{d}W_t, \qquad X_0 = x > 0.$$

Let $H_t = \exp((\mu - \sigma^2/2)t + \sigma W_t)$, so $\mathrm{d}H_t = \mu H_t \, \mathrm{d}t + \sigma H_t \, \mathrm{d}W_t$ and $H_0 = 1$.

(a) Compute $\mathrm{d}(H_t^{-1})$ and $\mathrm{d}[X, H^{-1}]_t$. Both X and H are driven by the same W.

(b) Apply the product rule to $X_t H_t^{-1}$. Check that its Brownian term and all drift terms proportional to X_t cancel.

(c) Integrate the remaining differential to obtain an explicit expression for X_t in terms of H and an ordinary time integral. Use it to prove $X_t > 0$.

(d) A proposed solution is $\widehat{X}_t = H_t(x + ct)$. Differentiate it and identify the inflow rate it actually implements. Explain why a contribution made at time s must grow by the factor H_t/H_s between s and t.

Part IV — Intensive Itô Formula Practice

Exercise 4.7 (Easy to medium — A time-dependent exponential martingale)**.** For $\lambda \in \mathbb{R}$, define

$$M_t^\lambda = \exp\Big(\lambda W_t - \tfrac{1}{2}\lambda^2 t\Big).$$

(a) Apply the time-dependent Itô formula to $f(t, x) = \exp(\lambda x - \lambda^2 t/2)$, and show that

$$\mathrm{d}M_t^\lambda = \lambda M_t^\lambda \, \mathrm{d}W_t.$$

Identify precisely which two drift contributions cancel.

(b) Prove that M^λ is a true martingale on every bounded horizon. You may compute $\mathbb{E}[(M_t^\lambda)^2]$ from the Gaussian moment-generating function.

(c) Let $a, b \in \mathbb{R}$. Determine all pairs (a, b) for which $\exp(aW_t + bt)$ has zero drift. Explain why the condition is about the quadratic variation of W, not ordinary calculus.

(d) Use conditional expectation to prove $\mathbb{E}[M_t^\lambda \mid \mathcal{F}_s] = M_s^\lambda$ for $s \le t$, directly from independent Brownian increments.

Exercise 4.8 (Medium — A backward equation is a martingale test)**.** Let $u \in C^{1,2}([0,T] \times \mathbb{R})$ have bounded u_x, and define $Z_t = u(t, W_t)$.

(a) Apply Itô's formula and show that Z is a true martingale whenever

$$u_t + \tfrac{1}{2} u_{xx} = 0.$$

Carefully state where boundedness of u_x is used.

(b) Let $g \in C^2(\mathbb{R})$ have bounded first and second derivatives, and set

$$u(t, x) = \mathbb{E}\big[g(x + W_{T-t})\big].$$

Explain, using independent increments and the martingale property from part (a), why $u(t, W_t) = \mathbb{E}[g(W_T) \mid \mathcal{F}_t]$.

(c) For $g(x) = \cos x$, compute u explicitly and verify the backward equation by differentiation.

(d) Explain why the same argument cannot apply directly to $g(x) = |x|$ at $x = 0$, and name the extension of Itô's formula that addresses this kink.

5 Week 5 — Self-Financing, Dividend Replication, and Greeks

Throughout, the money-market account is $B_t = e^{rt}$, with constant $r \in \mathbb{R}$. Under the historical measure $\mathbb{P}$, the ex-dividend stock price has dynamics

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}, \qquad \sigma > 0,$$

while one share pays the separate cash dividend $qS_t \,\mathrm{d}t$, with $q \geq 0$. Thus μ is the historical drift of the *price*; the shareholder's gain is $\mathrm{d}S_t + qS_t \,\mathrm{d}t$. Take $q = 0$ when an exercise specifies a non-dividend-paying stock.

Part I — Self-Financing and Dividend-Aware Replication

Exercise 5.1 (Medium — Derive self-financing from rebalancing)**.** First set $q = 0$. Trade at dates $0 = t_0 < \cdots < t_n = T$, holding Δ_i shares and β_i bank-account units on $(t_i, t_{i+1}]$. Let $X_i = \Delta_i S_{t_i} + \beta_i B_{t_i}$. Rebalancing takes place at the *observed* prices at t_{i+1}.

(a) Write the portfolio value just before and just after rebalancing at t_{i+1}. If no cash enters or leaves, show that

$$S_{t_{i+1}}(\Delta_{i+1} - \Delta_i) + B_{t_{i+1}}(\beta_{i+1} - \beta_i) = 0.$$

What pays for an increase in the stock holding?

(b) Deduce the exact wealth increment $X_{i+1} - X_i = \Delta_i(S_{t_{i+1}} - S_{t_i}) + \beta_i(B_{t_{i+1}} - B_{t_i})$. Explain why the old holdings appear.

(c) Pass to continuous time and state the self-financing gain equation

$$\mathrm{d}X_t = \Delta_t \,\mathrm{d}S_t + \beta_t \,\mathrm{d}B_t.$$

If Δ, β are continuous finite-variation processes, also apply the product rule to $X_t = \Delta_t S_t + \beta_t B_t$ and identify the equivalent rebalancing constraint.

(d) Restore the dividend yield q. Write the gain of one share and the self-financing equation. Explain why changing Δ without a compensating cash-account trade would require outside money.

Exercise 5.2 (Medium — Derive the PDE for a dividend-paying stock)**.** Let $V_t = v(t, S_t)$, where $v \in C^{1,2}([0,T) \times (0,\infty))$, and let a self-financing replicating portfolio hold Δ_t shares and β_t bank account units.

(a) Apply Itô's formula to V_t under the historical price dynamics specified above.

(b) Explain why the portfolio gain is

$$\mathrm{d}X_t = \Delta_t\big(\,\mathrm{d}S_t + qS_t \,\mathrm{d}t\big) + \beta_t \,\mathrm{d}B_t.$$

(c) Match the Brownian coefficients to identify Δ_t. Then use $\beta_t B_t = v(t, S_t) - \Delta_t S_t$ to derive

$$\boxed{v_t + (r-q)sv_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0.}$$

Locate precisely where the historical price drift μ cancels.

(d) Redo only the drift matching after incorrectly omitting the dividend gain $q\Delta_t S_t \,\mathrm{d}t$. Which term disappears from the PDE, and what economic cash flow has been lost?

Part II — PDE Applications and No-Dividend Greeks

Exercise 5.3 (Medium — Power payoffs reveal the roles of r and q)**.** For the terminal payoff $g(s) = s^p$, with $p \in \mathbb{R}$, seek a solution of the dividend-adjusted Black–Scholes PDE of the form $v(t, s) = A(t)s^p$.

(a) Derive the ODE for A, including $A(T) = 1$, and solve it.

(b) Check the special cases $p = 0$ and $p = 1$. Interpret the two prices as the value of one unit of cash at T and the value of one share delivered at T, respectively.

(c) Compute the replicating holdings $\Delta_t = v_s(t, S_t)$ and $\beta_t = (v - \Delta S)/B_t$.

(d) For $p > 1$, explain from the formula why curvature introduces a volatility contribution even though the physical drift μ is absent.

Exercise 5.4 (Medium — Derive the no-dividend call Greeks)**.** For this exercise set $q = 0$, $\tau = T - t > 0$, and use

$$C(t, s) = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_1 = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Write $\phi = \Phi'$, and hold the other model parameters fixed in each partial derivative.

(a) Show that $s\phi(d_1) = Ke^{-r\tau}\phi(d_2)$. Compute $\partial_s d_1$ and $\partial_s d_2$.

(b) Differentiate C with respect to s once and twice. Show every density term that cancels, and obtain delta and gamma.

(c) Compute $\partial_\sigma d_1$ and $\partial_\sigma d_2$, then derive vega. Derive rho by differentiating with respect to r.

(d) Differentiate first in τ, then use $\Theta = \partial_t C = -\partial_\tau C$ to derive calendar-time theta. State all five Greeks explicitly.

6 Week 6 — Densities, Girsanov, and Measure Changes

Throughout, all density processes are assumed adapted to the stated filtration. Whenever a stochastic exponential is used, distinguish the formal drift calculation from the condition that makes it a true martingale.

Part I — What a Density Really Encodes

Exercise 6.1 (Easy to medium — Reversing a change of measure)**.** Let $Z > 0$ satisfy $\mathbb{E}^{\mathbb{P}}[Z] = 1$, and define $\mathrm{d}\mathbb{Q} = Z\,\mathrm{d}\mathbb{P}$.

(a) Prove that

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} = \frac{1}{Z} \qquad \text{and} \qquad \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{Z}\right] = 1.$$

(b) Show that for every integrable Y,

$$\mathbb{E}^{\mathbb{P}}[Y] = \mathbb{E}^{\mathbb{Q}}\left[\frac{Y}{Z}\right].$$

Explain why the random variable itself has not changed.

(c) Use Jensen's inequality to prove

$$\mathbb{E}^{\mathbb{P}}[\log Z] \le 0.$$

Assuming the relevant integrability, prove also $\mathbb{E}^{\mathbb{Q}}[\log Z] \ge 0$.

(d) Characterize the equality case in both inequalities. What does it say about the relation between $\mathbb{P}$ and $\mathbb{Q}$?

Exercise 6.2 (Medium — A density process in a two-period tree)**.** Let

$$\Omega = \{UU, UD, DU, DD\},$$

where the first letter is observed at time 1 and the second at time 2. Under $\mathbb{P}$, all four paths have probability $1/4$. Under $\mathbb{Q}$, their probabilities in the displayed order are

$$\frac{3}{8}, \qquad \frac{1}{8}, \qquad \frac{1}{8}, \qquad \frac{3}{8}.$$

Let $\mathcal{F}_1$ reveal only the first letter, and let $Z_2 = \mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}$.

(a) Compute Z_2 on every terminal path and verify $\mathbb{E}^{\mathbb{P}}[Z_2] = 1$.

(b) Compute $Z_1 = \mathbb{E}^{\mathbb{P}}[Z_2 \mid \mathcal{F}_1]$. Explain why Z_1 is the density of the time-1 information under $\mathbb{Q}$ relative to $\mathbb{P}$.

(c) Compute under $\mathbb{Q}$ the conditional probability that the second letter is U, first given that the first letter is U, then given that it is D.

(d) Recover both answers from Bayes' formula

$$\mathbb{Q}(A \mid \mathcal{F}_1) = \frac{\mathbb{E}^{\mathbb{P}}[Z_2 \mathbf{1}_A \mid \mathcal{F}_1]}{Z_1}.$$

Does the second step remain independent of the first under $\mathbb{Q}$?

Part II — Tilting Beyond the Classroom Examples

Exercise 6.3 (Medium — Tilting a correlated Gaussian vector)**.** Under $\mathbb{P}$, let (X, Y) be centered jointly Gaussian with

$$\mathrm{Var}(X) = \sigma_X^2, \qquad \mathrm{Var}(Y) = \sigma_Y^2, \qquad \mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y.$$

For $\theta \in \mathbb{R}$, define

$$Z = \exp\left(\theta X - \frac{1}{2}\theta^2\sigma_X^2\right), \qquad \mathrm{d}\mathbb{Q} = Z\ \mathrm{d}\mathbb{P}.$$

(a) Verify that $\mathbb{E}^{\mathbb{P}}[Z] = 1$.

(b) Use the joint moment-generating function to compute $\mathbb{E}^{\mathbb{Q}}[e^{uX+vY}]$.

(c) Identify the mean vector and covariance matrix of (X, Y) under $\mathbb{Q}$. Explain why tilting with X also moves the mean of Y when $\rho \neq 0$.

(d) Determine for which ρ the law of Y is unchanged by this measure change.

Exercise 6.4 (Medium — An adapted tilt made safe by stopping)**.** Let

$$\tau_a = \inf\{t \geq 0 : |W_t| \geq a\} \wedge T, \qquad \phi_t = \theta W_t \mathbf{1}_{\{t \leq \tau_a\}},$$

where $a, T > 0$ and $\theta \in \mathbb{R}$. Define

$$Z_t = \exp\left(\int_0^t \phi_s\, \mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2\, \mathrm{d}s\right).$$

(a) Show that $|\phi_t| \leq |\theta|a$, and verify Novikov's condition. Conclude that Z is a true martingale.

(b) Define $\mathbb{Q}$ by Z_T. Identify the $\mathbb{Q}$-Brownian motion B and write the $\mathbb{Q}$-dynamics of W.

(c) Describe the drift of W before and after τ_a. For $\theta > 0$, does the pre-stopping drift push the process toward or away from zero?

(d) Explain why stopping is doing genuine mathematical work here rather than merely shortening the notation.

Part III — Designing and Testing a Measure Change

Exercise 6.5 (Medium — Changing one mean-reversion model into another)**.** Under $\mathbb{P}$, suppose

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \kappa, \sigma > 0.$$

We would like X to satisfy under $\mathbb{Q}$

$$\mathrm{d}X_t = \widetilde{\kappa}(\widetilde{m} - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \widetilde{\kappa} > 0.$$

(a) Find the candidate process λ_t in $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \lambda_s\,\mathrm{d}s$.

(b) Write the corresponding candidate density process, with the correct sign.

(c) Explain why the volatility must remain σ. Could an equivalent Girsanov change turn it into a different constant $\widetilde{\sigma} \neq \sigma$?

(d) The candidate λ_t is generally unbounded. Explain why the algebraic drift match does not by itself prove that $\mathbb{Q}$ exists, and describe how localization by $\tau_n = \inf\{t : |X_t| \geq n\} \wedge T$ helps.

Exercise 6.6 (Medium — Can one Brownian shift price two stocks?)**.** Two traded stocks are driven by the same Brownian motion:

$$\frac{\mathrm{d}S_t^1}{S_t^1} = \mu_1 \,\mathrm{d}t + \sigma_1 \,\mathrm{d}W_t, \qquad \frac{\mathrm{d}S_t^2}{S_t^2} = \mu_2 \,\mathrm{d}t + \sigma_2 \,\mathrm{d}W_t,$$

where $\sigma_1, \sigma_2 \neq 0$, and the short rate is r.

(a) If $W_t^{\mathbb{Q}} = W_t + \lambda t$, derive the condition on λ that makes each discounted stock driftless.

(b) Show that a common equivalent martingale measure of this form can exist only if

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2}.$$

Interpret this equality economically.

(c) When the equality holds, identify the density process and explain why the Brownian risk premium is unique.

(d) Suppose the equality fails. Find dollar exposures x_1, x_2 in the two stocks such that $x_1\sigma_1 + x_2\sigma_2 = 0$, while $x_1(\mu_1 - r) + x_2(\mu_2 - r) \neq 0$. Explain why this is an instantaneous arbitrage signal and how the sign of the position can be chosen.

7 Week 7 — Martingale Pricing, Hedging, and Market Structure

Throughout, $\mathbb{Q}$ denotes a martingale measure for the stated numeraire. Every time a local martingale is used inside a conditional expectation, state the integrability or admissibility assumption that makes it a true martingale.

Part I — Discounted Gains and Conditional Valuation

Exercise 7.1 (Medium — Discounted gains when one asset pays dividends)**.** Let

$$B_t = \exp\left(\int_0^t r_u \,\mathrm{d}u\right), \qquad D_t = B_t^{-1}.$$

Two risky assets are traded. Asset S^1 pays no dividends, while one share of S^2 pays the continuous cashflow $q_t S_t^2 \,\mathrm{d}t$. A predictable self-financing strategy has holdings $(\phi_t^0, \phi_t^1, \phi_t^2)$ and wealth

$$X_t = \phi_t^0 B_t + \phi_t^1 S_t^1 + \phi_t^2 S_t^2.$$

(a) Write the correct self-financing gain equation for X.

(b) Prove the discounted gain identity

$$\mathrm{d}(D_t X_t) = \phi_t^1 \,\mathrm{d}(D_t S_t^1) + \phi_t^2 \left(\mathrm{d}(D_t S_t^2) + D_t q_t S_t^2 \,\mathrm{d}t\right).$$

(c) Suppose that, under $\mathbb{Q}$, the discounted gain of each risky asset is a local martingale. Explain why DX is then a local martingale. Which additional issue must be settled before writing

$$D_t X_t = \mathbb{E}^{\mathbb{Q}}[D_T X_T \mid \mathcal{F}_t]?$$

(d) A student claims that DS^2 itself must be a martingale. Explain precisely why this is generally false when $q \neq 0$.

Exercise 7.2 (Medium — A payoff determined at an intermediate date)**.** Assume r is constant and $B_t = e^{rt}$. Fix $0 < u < T$, and let $Y \in L^1(\mathbb{Q})$ be $\mathcal{F}_u$-measurable. At time T, a contract pays

$$H = e^{r(T-u)} Y.$$

(a) For $t \leq u$, derive the price process in terms of $\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t]$.

(b) For $u \leq t \leq T$, show that the price is $V_t = e^{r(t-u)} Y$.

(c) Check that the two expressions agree at $t = u$, and interpret the contract as an operation performed at time u.

(d) Prove directly that $e^{-rt} V_t$ is a martingale on $[0, T]$. Why is the tower property the key step before u?

Part II — New Martingale Representations and Hedges

Exercise 7.3 (Medium — An exponential terminal claim)**.** Let W be a $\mathbb{Q}$-Brownian motion, let $\theta \in \mathbb{R}$, and assume zero interest for simplicity. Consider

$$H = \exp\left(\theta W_T - \frac{1}{2}\theta^2 T\right).$$

(a) Compute $M_t = \mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t]$ using independent Brownian increments.

(b) Apply Itô's formula to identify the process ψ_t in

$$M_t = M_0 + \int_0^t \psi_s \, \mathrm{d}W_s.$$

(c) Suppose the discounted traded stock satisfies

$$\mathrm{d}\bar{S}_t = \sigma_t \bar{S}_t \, \mathrm{d}W_t, \qquad \sigma_t \bar{S}_t \neq 0.$$

Find the stock holding that replicates H.

(d) What fails on a set where $\sigma_t \bar{S}_t = 0$? Distinguish the existence of the Brownian representation from its implementation by traded assets.

Exercise 7.4 (Medium — The Brownian area as a hedgeable claim)**.** Let W be a $\mathbb{Q}$-Brownian motion, $r = 0$, and

$$H = \int_0^T W_s \, \mathrm{d}s.$$

(a) Use stochastic Fubini or deterministic integration by parts to prove

$$H = \int_0^T (T - u) \, \mathrm{d}W_u.$$

(b) Compute $M_t = \mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t]$ in terms of $\int_0^t W_s \, \mathrm{d}s$ and W_t.

(c) Apply Itô's formula to your answer and recover the representation coefficient from part (a).

(d) Compute $\mathrm{Var}^{\mathbb{Q}}(H)$ in two different ways.

(e) If $\mathrm{d}\bar{S}_t = \gamma_t \, \mathrm{d}W_t$ with $\gamma_t \neq 0$, find the replicating stock holding. Why does the hedge decrease as maturity approaches?

Part III — Spanning, Redundancy, and Claim-Specific Completeness

Exercise 7.5 (Medium — Complete risk spanning with redundant assets)**.** Three discounted risky assets are driven by two $\mathbb{Q}$-Brownian factors:

$$\mathrm{d}\bar{S}_t = \Gamma \, \mathrm{d}W_t, \qquad \Gamma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

A discounted claim has MRT exposure $\psi_t = (a_t, b_t)^\top$.

(a) Show that every $\psi_t \in \mathbb{R}^2$ can be hedged.

(b) Find the full family of holdings $\phi_t = (\phi_t^1, \phi_t^2, \phi_t^3)^\top$ satisfying $\Gamma^\top \phi_t = \psi_t$.

(c) Among this family, find the hedge with the smallest Euclidean norm.

(d) Identify a nonzero zero-exposure portfolio. Explain why nonuniqueness of holdings here does not imply market incompleteness or price ambiguity.

Exercise 7.6 (Medium to challenging — Which claims remain replicable?)**.** The filtration is generated by two independent $\mathbb{Q}$-Brownian motions (W^1, W^2). The only discounted risky asset satisfies

$$\mathrm{d}\bar{S}_t = \sigma_t \bar{S}_t \,\mathrm{d}W_t^1, \qquad \sigma_t \bar{S}_t \neq 0.$$

Let deterministic $f, g \in L^2(0,T)$, and consider

$$H_{f,g} = c + \int_0^T f_t \,\mathrm{d}W_t^1 + \int_0^T g_t \,\mathrm{d}W_t^2.$$

(a) Find the martingale $M_t = \mathbb{E}^{\mathbb{Q}}[H_{f,g} \mid \mathcal{F}_t]$ and its two-dimensional MRT exposure.

(b) Prove that $H_{f,g}$ is replicable using the stock and cash if and only if $g = 0$ almost everywhere.

(c) When $g = 0$, find the stock and cash holdings that replicate the claim.

(d) Explain why the market can be incomplete even though every claim in the subfamily $\{H_{f,0}\}$ has a unique replication price.

Part IV — Forward Pricing and Unspanned Risk

Exercise 7.7 (Medium — A forward with time-dependent deterministic rates)**.** Assume under $\mathbb{Q}$

$$\frac{\mathrm{d}S_t}{S_t} = r(t)\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t^{\mathbb{Q}},$$

where r, σ are deterministic and square-integrable on $[0,T]$. Let $B_t = \exp(\int_0^t r(u)\,\mathrm{d}u)$.

(a) Apply Itô's formula to $\log S_u$, integrate from t to T, and exponentiate to obtain S_T. Then give the conditional mean and variance of $\log(S_T/S_t)$ given $\mathcal{F}_t$. Why is its stochastic-integral term independent of $\mathcal{F}_t$?

(b) Prove directly that S_t/B_t is a true $\mathbb{Q}$-martingale. State the integrability fact you use.

(c) Price the forward-style payoff $H = S_T - K$ by conditional expectation and show

$$V_t = S_t - K \exp\left(-\int_t^T r(u)\,\mathrm{d}u\right).$$

(d) Determine the replicating stock holding and explain why the volatility function disappears from the price but not from the distribution of the payoff.

Exercise 7.8 (Medium to challenging — An unspanned Brownian risk)**.** Let W^1, W^2 be independent Brownian motions under $\mathbb{P}$. The only risky traded asset satisfies

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^1, \qquad \sigma > 0,$$

and the short rate is r. Set $\lambda = (\mu - r)/\sigma$. For each $\eta \in \mathbb{R}$, define

$$Z_T^\eta = \exp\left(-\lambda W_T^1 - \eta W_T^2 - \frac{1}{2}(\lambda^2 + \eta^2)T\right).$$

(a) Verify that Z^η is a density process and identify the two Brownian motions under the associated measure $\mathbb{Q}^\eta$.

(b) Show that $e^{-rt}S_t$ is a martingale under every $\mathbb{Q}^\eta$. Why does the traded stock impose no restriction on η?

(c) Consider the nontraded claim $H = (W_T^2)^2$. Compute $\mathbb{E}^{\mathbb{Q}^\eta}[H]$ and the corresponding discounted expectation price.

(d) Explain why the family of different prices does not contradict no-arbitrage. What missing traded risk would be needed to make the claim replicable?

Part V — Path Dependence

Exercise 7.9 (Medium — A weighted and floating-strike Asian family)**.** Under $\mathbb{Q}$, assume

$$\frac{\mathrm{d}S_t}{S_t} = (r-q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t,$$

with constant r, q, σ. Let $w : [0,T] \to [0,\infty)$ be deterministic and define

$$A_t = \int_0^t w(u)S_u\,\mathrm{d}u.$$

Consider the payoff

$$H = (\alpha S_T + \beta A_T - K)^+.$$

(a) Show that (S_t, A_t) is a Markov state and write its dynamics.

(b) Assuming $V_t = v(t, S_t, A_t)$, derive the pricing PDE and terminal condition.

(c) Explain why there are no v_{aa} or v_{sa} terms.

(d) Identify the stock holding in the replicating hedge.

(e) Recover as special cases a weighted fixed-strike Asian call and a floating-strike Asian call. State the corresponding choices of α, β, K.

Part VI — A Finite-Market Test of the FTAP

Exercise 7.10 (Challenging — EMM interval, replication, and superhedging)**.** Consider a one-period market with zero interest and three states $\omega_1, \omega_2, \omega_3$, each having positive physical probability. The bond satisfies $B_0 = B_1 = 1$, while

$$S_0 = 2, \qquad S_1(\omega_1) = 1, \quad S_1(\omega_2) = 2, \quad S_1(\omega_3) = 4.$$

(a) Find every equivalent martingale measure $\mathbb{Q}^c$. Express its three state probabilities in terms of one parameter c and determine the admissible interval for c.

(b) Use the FTAP to explain why the market is arbitrage-free but incomplete.

(c) Show that a payoff $h = (h_1, h_2, h_3)$ is replicable if and only if

$$h_3 - 3h_2 + 2h_1 = 0.$$

(d) Let $H = (0,1,0)$, the claim paying only in the middle state. Compute its expectation under every $\mathbb{Q}^c$ and determine the full no-arbitrage price interval when H is added as a traded claim.

(e) Find the cheapest stock–bond superhedge and the most expensive stock–bond subhedge for H. Relate their costs to the endpoints of the price interval.

8 Week 8 — No-Arbitrage Principles and the Fundamental Theorems of Asset Pricing

Throughout, $B_t = e^{rt}$ denotes the money-market account under a constant short rate r, and $P_t(T)$ denotes the time-t price of a zero-coupon bond paying 1 at T. In one-period examples, R denotes the bond's gross return over the period.

Part I — Dominance and the Law of One Price

Exercise 8.1 (Easy — Turning a paid-to-enter trade into a free-and-sometimes-profitable one)**.** A one-period market has two states ω_1, ω_2 and a bond with price $S_0^0 = 1$ and payoff $R = 1.08$ in both states. A strategy θ has

$$V_0(\theta) = -4, \qquad V_T(\theta, \omega_1) = 2, \qquad V_T(\theta, \omega_2) = 0.$$

(a) Check that θ is a type-(ii) arbitrage in the sense used in lecture.

(b) Augment θ with the correct number of units of the bond so that the new strategy costs exactly 0 today. Compute its terminal value in each state and confirm it is a type-(i) arbitrage.

(c) Explain, in one sentence and without redoing the arithmetic, why the same construction works for *any* strategy with $V_0(\theta) < 0$ and $V_T(\theta, \cdot) \geq 0$.

Exercise 8.2 (Easy to medium — Marking an existing forward to market)**.** A forward contract on a non-dividend-paying stock was entered at some past date with a fixed delivery price K — not necessarily the fair forward price at inception. The contract still pays $S_T - K$ at maturity T.

(a) Build the static replicating portfolio for this payoff at a current time $t < T$, and use the law of one price to find its current value V_t in terms of S_t, K, r, and $T - t$.

(b) Check that your formula gives $V_t = 0$ when K equals the fair forward price $K_t = S_t e^{r(T-t)}$ quoted for a *new* contract entered at t.

(c) *Numerical.* A long forward was entered when $S_{t_0} = 100$, $r = 4\%$, with six months to maturity and delivery price $K = 102$. Three months later, with three months left to maturity, the stock trades at $S_t = 105$. Find the current value of the long forward position.

Part II — Completeness and the Second Fundamental Theorem

Exercise 8.3 (Medium — Completing a four-state market may take two new assets)**.** One period, four states. A bond pays 1 in every state. A stock with $S_0 = 100$ pays $(70, 90, 110, 130)$ across the four states. A call struck at $K = 105$ pays $(0, 0, 5, 25)$.

(a) Explain, without any computation, why three traded assets can never span four states.

(b) Using the states with payoffs $(70, 90, 130)$ — i.e., omitting the third state — form the 3×3 matrix of bond, stock, and call payoffs and compute its determinant. Conclude that the three assets are linearly independent.

(c) A second call, struck at $K = 115$, pays $(0, 0, 0, 15)$. Form the 4×4 payoff matrix of all four states and all four assets (bond, stock, call at 105, call at 115), and compute its determinant.

(d) Conclude whether the market is complete once the second call is added, and relate your answer to the Second Fundamental Theorem.

Part III — Superhedging and Price Bounds

Exercise 8.4 (Easy to medium — Bounding a worst-of-two-stocks claim)**.** Let $H = \min(S_T^1, S_T^2)$, the mirror image of the best-of-two-stocks claim from lecture.

(a) Show $\min(a, b) \leq a$ and $\min(a, b) \leq b$ for $a, b \geq 0$. Conclude that holding either stock alone *superhedges* H, and state the resulting upper bound on $V_t(H)$.

(b) Explain why holding cash alone ($Y_T \equiv 0$) is a valid but very loose subhedge. Why does a materially tighter lower bound generally require a claim on the *spread* $S_T^1 - S_T^2$, rather than a position in the two stocks alone?

(c) *Numerical.* With $S_t^1 = 100$ and $S_t^2 = 95$, state the resulting price corridor for H.